

Chapter 7 - Class A Subnetting, Subnet Mask & VLSM

1. What is a Class A IP Address?

A Class A IP address is designed for very large networks .

Class A Basics

- Range: **1.0.0.0 – 126.255.255.255**
- **1st Octet** → Network portion
- **Last 3 Octets** → Host portion
- Default CIDR: **/8**

Daily Life Example:

- A multinational company  with offices across countries
- One main network with thousands of devices

2. Class A Subnetting Rules

◆ Finding Number of Networks

- Count how many **bits are borrowed** from host part

Formula:

Number of Networks = 2^n

(n = borrowed bits)

◆ Finding IP Addresses per Network

- Count **remaining host bits**

Formula:

Total IPs per Network = 2^n

◆ Finding Usable Hosts

Usable Hosts = Total IPs – 2

? Why subtract 2?

- 1 Network ID
- 1 Broadcast ID



3. Class A Subnetting Example: 10.0.0.0/28

◆ Step 1: Identify Class

- 10.x.x.x → Class A
- Default mask = /8

◆ Step 2: Borrowed Bits

$28 - 8 = 20$ bits borrowed

◆ Step 3: Number of Networks

$2^{20} = 1,048,576$ networks

◆ Step 4: Host Bits & Hosts

- Host bits = $32 - 28 = 4$ bits

Total IPs = $2^4 = 16$

Usable Hosts = 14

◆ Step 5: Subnet Mask

/28 = 255.255.255.240



4. What is a Subnet Mask?

A **Subnet Mask** is a **32-bit number** used to divide an IP address into:

- Network portion
- Host portion



Example

- IP: 192.168.10.0
- Subnet Mask: 255.255.255.0
- CIDR: /24



First 24 bits = Network



Last 8 bits = Host



5. Why Subnet Mask is Used?

- Helps routers identify the correct network 🚦
- Divides large networks into smaller subnets 🧩
- Controls number of usable hosts



Daily Life Example:

- Big office divided into HR, IT, Sales networks 🏢



6. Default Subnet Masks

Class Subnet Mask

Class A 255.0.0.0

Class B 255.255.0.0

Class C 255.255.255.0

7. Subnet Mask to CIDR Example

Example:

Subnet Mask: 255.255.255.248

- Binary last octet: **11111000**
- Network bits = **29**

👉 CIDR = **/29**

👉 Host bits = **5 bits used**

8. Subnetting Cheat Sheet

Group Size	Subnet Mask	CIDR
128	255.255.255.128	/25
64	255.255.255.192	/26
32	255.255.255.224	/27
16	255.255.255.240	/28
8	255.255.255.248	/29
4	255.255.255.252	/30
2	255.255.255.254	/31
1	255.255.255.255	/32

9. /30 Subnet Mask

/30 = 255.255.255.252

👉 Commonly used for **router-to-router links** 



10. Checking IP Address Using CMD

- Use command:

ipconfig



Shows:

- Private IP Address
- Subnet Mask
- Default Gateway



11. What is VLSM (Variable Length Subnet Mask)?

VLSM allows different subnet masks for different networks 🎯.



Why VLSM?

- Saves IP addresses
- Efficient network design



12. VLSM Example Scenario



Given Network

192.168.10.0/26

Total IPs = 64

Usable = 62



Requirements

Team	Hosts Needed
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Team A	60
--------	----

Team B	30
--------	----

Team C	12
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Team Hosts Needed

Team D 6

13. Power of 2 Table (2^7 to 2^0)

Power Value

2^7 128

2^6 64

2^5 32

2^4 16

2^3 8

2^2 4

2^1 2

2^0 1

14. Step 1: Arrange Requirements (Largest to Smallest)

60 → 30 → 12 → 6

15. Step 2: Find Required Subnet Mask

Hosts Needed Nearest Power of 2 Total IPs Subnet Mask CIDR

60 64 64 255.255.255.192 /26

30 32 32 255.255.255.224 /27

12 16 16 255.255.255.240 /28

6 8 8 255.255.255.248 /29



16. Step 3: Assign Networks (Largest to Smallest)

◆ Base Network: 192.168.10.0

Team	Network ID	IP Range	Broadcast
Team A (60)	192.168.10.0/26	.1 – .62	.63
Team B (30)	192.168.10.64/27	.65 – .94	.95
Team C (12)	192.168.10.96/28	.97 – .110	.111
Team D (6)	192.168.10.112/29	.113 – .118	.119



Key Takeaways

- Class A default mask = /8
 - Subnet mask defines network & host bits
 - VLSM saves IP addresses
 - Always allocate largest network first
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◆ 1 Chapter Summary

This chapter explains **advanced subnetting concepts** used in **very large enterprise networks** 🌐.

You learned:

- What a **Class A IP address** is and where it is used
- How **Class A subnetting** works using CIDR
- What a **Subnet Mask** is and why it is important
- Default subnet masks of Class A, B, and C
- Conversion between **Subnet Mask and CIDR**
- Practical use of **/30 subnet** for router links
- How to check IP details using **CMD (ipconfig)**
- Most importantly, **VLSM (Variable Length Subnet Mask)** for efficient IP utilization

👉 This chapter moves you from **basic subnetting** to **real-world network design** 📁 🌐

◆ 2 Chapter Conclusion 🎯

Class A subnetting and VLSM are **core skills for network engineers** working with:

- Large enterprises 🏢
- ISPs 🌐
- Data centers 💻

Key understanding from this chapter:

- **Subnet Mask** decides how a network is divided
- **CIDR** removes class limitations
- **VLSM** prevents IP wastage and optimizes networks

- Always assign **largest subnet first** in VLSM

💡 If you can explain VLSM confidently, an interviewer instantly knows:

“This candidate understands real network planning.” 🔥 😎

◆ 3 Detailed Mind Map

Lecture 7: Class A Subnetting, Subnet Mask & VLSM

|

└─ Class A IP Address

|

└─ Range: 1.0.0.0 – 126.255.255.255

|

└─ Default Mask: /8

|

└─ 1st Octet = Network

|

└─ Last 3 Octets = Host

|

└─ Class A Subnetting

|

└─ Borrowed bits from host

|

└─ Networks = 2^n

|

└─ Hosts = $2^n - 2$

|

└─ Example: 10.0.0.0/28

|

└─ Subnet Mask

|

└─ 32-bit number

|

└─ Divides network & host

|

└─ Default masks (A/B/C)

|

└─ CIDR representation

|

- └─ Subnet Mask ↔ CIDR

- └─ Binary conversion

- └─ Count network bits

- └─ Example: 255.255.255.248 → /29

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- └─ Special Subnets

- └─ /30 for router links

- └─ /32 for single host

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- └─ CMD Networking

- └─ ipconfig

- └─ IP address

- └─ Subnet mask

- └─ Default gateway

-

- └─ VLSM

- └─ Variable subnet sizes

- └─ Efficient IP usage

- └─ Largest-first allocation

- └─ Real-world design

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- └─ VLSM Example

- └─ Requirement analysis

- └─ Power of 2 selection

- └─ CIDR assignment

- └─ Network allocation

◆ Q & A

? Q1. What is a Class A IP address?

Answer:

A Class A IP address is designed for **very large networks**.

Its range is **1.0.0.0 to 126.255.255.255**, and its default subnet mask is **/8**.

? Q2. How many network and host bits are there in Class A?

Answer:

- Network bits: 8
- Host bits: 24

This allows **millions of hosts** in a single network .

? Q3. What is the default CIDR of Class A?

Answer:

The default CIDR of Class A is **/8**.

? Q4. Explain Class A subnetting in simple terms.

Answer:

Class A subnetting means **borrowing bits from the host portion** to create multiple smaller networks while reducing hosts per network.

? Q5. In 10.0.0.0/28, how many bits are borrowed?

Answer:

Borrowed bits = $28 - 8 = 20$ bits

? Q6. How many networks are created in 10.0.0.0/28?

Answer:

Networks = $2^{20} = 1,048,576$ networks

? Q7. How many usable hosts are in /28?

Answer:

Host bits = 4

Total IPs = 16

Usable hosts = **14**

? Q8. What is a Subnet Mask?

Answer:

A subnet mask is a **32-bit number** used to separate:

- Network portion
- Host portion

It helps routers decide where to send packets 🚦 .

? Q9. Why is subnet mask important?

Answer:

Subnet mask:

- Divides large networks
 - Improves security 🛡️
 - Reduces traffic 🚦
 - Controls number of hosts
-

? Q10. What are default subnet masks of Class A, B, and C?

Answer:

- Class A → 255.0.0.0
 - Class B → 255.255.0.0
 - Class C → 255.255.255.0
-

? Q11. How do you convert subnet mask to CIDR?

Answer:

Convert subnet mask to binary and **count the number of 1s**.

Example:

255.255.255.248 → 11111000 → /29

? Q12. Why is /30 subnet commonly used?

Answer:

/30 provides **2 usable IPs**, which is perfect for **router-to-router point-to-point links** 🔗 .

? Q13. What command is used to check IP details in Windows?

Answer:

The command is:

ipconfig

It shows IP address, subnet mask, and default gateway 💻 .

? Q14. What is VLSM?

Answer:

VLSM (Variable Length Subnet Mask) allows **different subnet sizes** within the same network to **save IP addresses** 🎯 .

? Q15. Why is VLSM better than FLSM?

Answer:

Because VLSM:

- Avoids IP wastage
 - Allocates IPs based on actual requirement
 - Is used in real-world networks 🌐
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? Q16. What is the first step in VLSM?

Answer:

Arrange host requirements from **largest to smallest** 📄.

? Q17. Why do we assign the largest subnet first in VLSM?

Answer:

Because large networks need continuous IP ranges.

If assigned later, IP fragmentation can occur ❌.

? Q18. Explain the VLSM example in brief.

Answer:

From 192.168.10.0/26:

- Team A → /26
- Team B → /27
- Team C → /28
- Team D → /29

This ensures **zero IP wastage** ✔️.

? Q19. What happens if VLSM is not used?

Answer:

Without VLSM:

- IP addresses are wasted
 - Network design becomes inefficient
 - Scaling becomes difficult 🚨
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? Q20. Real-life analogy of VLSM?

Answer:

Different departments in a company get **different-sized office spaces** based on team size 🏢.